

- 8 ai Using Fleming's left hand rule, the magnetic force on the wire is upwards. **B1**
By Newton's third law, the force on the magnet is downwards. **B1**
- aii $F = BIL \sin \theta$
 $= 3.7 \times 10^{-3} \times 5.1 \times 8.5 \times 10^{-2} \times \sin 90^\circ$ **C1**
 $= 1.6 \times 10^{-3} \text{ N}$ **A1**
- aiii New length of wire in magnetic field = $\frac{8.5 \times 10^{-2}}{\sin 60^\circ}$
- $F = BIL \sin \theta$
 $= 3.7 \times 10^{-3} \times 5.1 \times \frac{8.5 \times 10^{-2}}{\sin 60^\circ} \times \sin 60^\circ$ **M1**
 $= 1.6 \times 10^{-3} \text{ N}$ **A1**
- aiv Since $F = BIL \sin \theta$ or F is directly proportional to its current, an alternating current leads to a sinusoidal variation of F with time (at least 2 cycles). **B1**
- Maximum force (amplitude) = $\sqrt{2} \times 1.6 \times 10^{-3} = 2.3 \times 10^{-3} \text{ N}$ **B1**
- $T = \frac{1}{f} = \frac{1}{20} = 0.050 \text{ s}$ **B1**
- bi A magnetic field is region where there is a force **M1**
experienced by a current-carrying conductor/moving charge/permanent magnet. **A1**
- bii an arrow on the axis of the solenoid pointing downwards labelled P **B1**
- biii 1. Lenz's law states that the direction of the induced e.m.f (or current) produced **M1**
is such as to cause effects to oppose the change producing it. **A1**
- biii 2. As the current in the solenoid is switched on, the (downwards) magnetic field in solenoid is increasing. **B1**
The small coil experiences an increasing magnetic flux linkage, hence it induces a magnetic field in the coil in the opposite direction to oppose the increase. **B1**

The arrow of induced magnetic field is inside or just above small coil pointing in opposite direction to P is upwards.

B1

biv $\Delta NBA \cos \theta = 75 \times (1.4 \times 10^{-3} \times 2) \times 7.0 \times 10^{-4}$

C1

$$E = \frac{d\phi}{dt} \approx \frac{\Delta NBA \cos \theta}{\Delta t}$$

C1

$$= \frac{75 \times (1.4 \times 10^{-3} \times 2) \times 7.0 \times 10^{-4}}{0.12}$$

A1

$$= 1.2 \times 10^{-3} \text{ V}$$